1 **(a)** Evidence of using $\dfrac{\text{change in } y}{\text{change in } x}$ with two correct points from the graph, e.g., $\dfrac{4-(-2)}{3-0}$ **[M1]**

2 **[A1]**

(b) $(0, -2)$ **[B1]**

[Total Marks: 3]

2 **(a)** $\dfrac{(a+6)-3a}{4-(-2)}$ **[M1]**

$\dfrac{6-2a}{6}$ **[M1]**

$\dfrac{3-a}{3}$ or $1-\dfrac{a}{3}$ **[A1]**

(b) $\dfrac{3-a}{3} = 3$ oe **[M1]**

$a = -6$ **[A1]**

(c) $P = (-2, -18)$ and $Q = (4, 0)$ **[M1]**

$\left(\dfrac{-2+4}{2}, \dfrac{-18+0}{2}\right)$ **[M1]**

$(1, -9)$ **[A1]**

[Total Marks: 8]

3 **(a)** Uses gradient formula: $\dfrac{-4-8}{4-(-2)}$ or $\dfrac{8-(-4)}{-2-4}$ **[M1]**

-2 **[A1]**

(b) Identifies coordinates of two clear points on the line, e.g., $(0, 1)$ and $(4, 4)$ **[M1]**

$\dfrac{3}{4}$ or 0.75 **[A1]**

[Total Marks: 4]

4 **(a)** Calculates the gradient of AB as $\dfrac{9-3}{4-1} = 2$ and equates to gradient of BC or AC, e.g.,

$\dfrac{y-9}{8-4} = 2$ **[M1]**

$y = 17$ **[A1]**

(b) Sets up the gradient equation: $\dfrac{(p^2-p-3)-3}{p-1} = 2$ **[M1]**

Simplifies to form a 3-term quadratic equation: $p^2 - p - 6 = 2(p-1)$

$\mathrm{p}^2 - 3p - 4 = 0$ [M1]

Factorises $(p-4)(p+1) = 0$ or uses the quadratic formula correctly **[M1]**

$p = 4$ and $p = -1$ **[A1]**

(c) Sets up the gradient equation: $\dfrac{12-m}{(m+1)-3} = m$ **[M1]**

Multiplies by the denominator: $12 - m = m(m-2)$ **[M1]**

Rearranges into a 3-term quadratic: $m^2 - m - 12 = 0$ **[M1]**

$m = 4$ and $m = -3$ **[A1]**

[Total Marks: 10]

5 (a) $\dfrac{-9-5}{4-(-3)}$ or $\dfrac{5-(-9)}{-3-4}$ **[M1]**

-2 **[A1]**

(b) $\dfrac{1-5}{k-(-3)} = -2$ oe **[M1]**

$-4 = -2(k+3)$ or $-2k - 6 = -4$ **[M1]**

$k = -1$ **[A1]**

[Total Marks: 5]

6 (a) Forms a correct equation for the gradient: $\dfrac{5-2k}{2-(k-3)} = -3$ OR $\dfrac{2k-5}{k-3-2} = -3$ **[M1]**

Correctly simplifies the denominator and equates: $5 - 2k = -3(5-k)$ **[M1]**

$k = 4$ **[A1]**

(b) Evaluates coordinates of P as $(1, 8)$ and substitutes into midpoint formula: $\left(\dfrac{1+2}{2}, \dfrac{8+5}{2}\right)$ **[M1]**

$(1.5, 6.5)$ **[A1]**

[Total Marks: 5]